

Development of THz Cavities

This section presents the design, calculation, and fabrication of the cavities, including the process of realizing the electromagnetic fields confinement trap.

Design of THz Cavities

One key component in this project is the development of a resonant cavity with low mode volume and high-quality factor, capable of confining and storing electromagnetic fields with minimal losses. These properties support long photon lifetimes and enhanced light–matter interactions needed for effective operation in the dispersive regime, enabling on-demand, non-destructive readout of the qubit state, together with controlled manipulation of qubit states.

We chose a CPW transmission line as the resonator due to its favourable properties. Its small mode volume produces large vacuum electric fields from zero-point fluctuations, and it can exhibit high internal quality factors. These properties make CPW resonators strong candidate for this application, in particular for coherent single-qubit control, dispersive readout, and qubit–qubit coupling via a resonator bus [38].

Their compatibility with our fabrication process and potential extension to higher frequencies further motivate this choice [42]. We also use a gap capacitor as the coupling element for its fabrication compatibility and ease of integration.

The device is fabricated on a 20×20 mm chip divided into four sections, each containing three cavities and one reference cavity, giving 16 cavities in total.

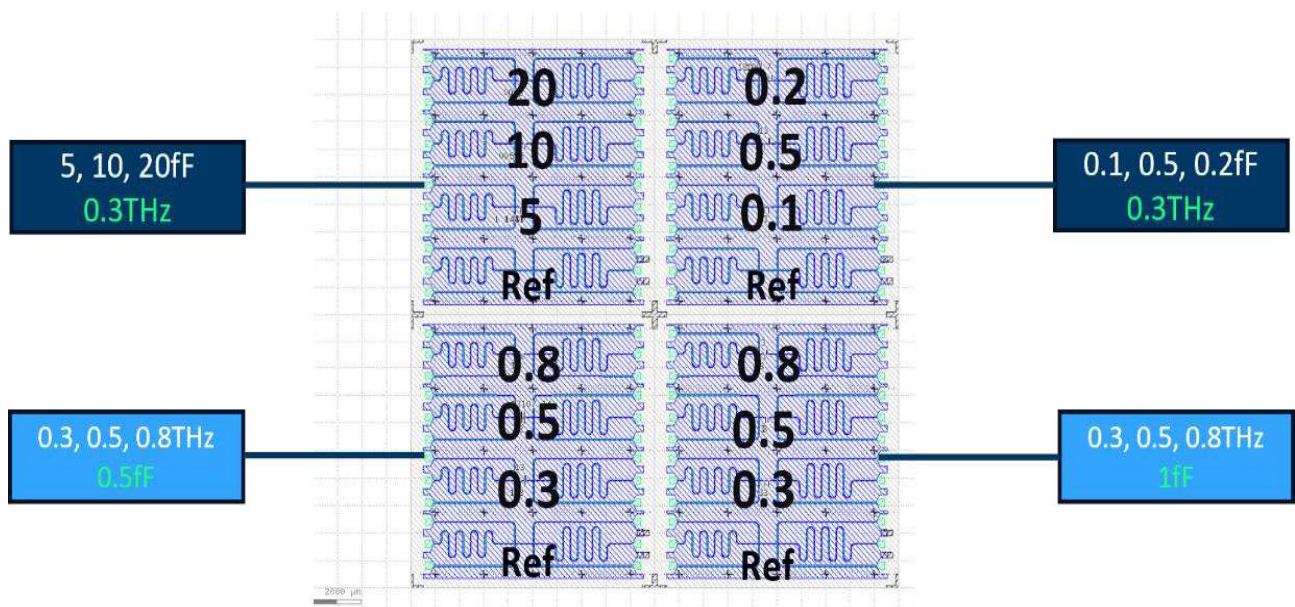


Figure 11. Summary of cavities in a 20x20mm sample size

Coupling Capacitance (Gap Capacitor) for THz Cavities

For this calculation, a quasi-static analytical model reported by Simons [39] was used to determine the coupling capacitance values of the different cavities for a CPW with centre conductor width $W = 10 \mu\text{m}$. These capacitances are subsequently employed to compute the corresponding resonance frequencies.

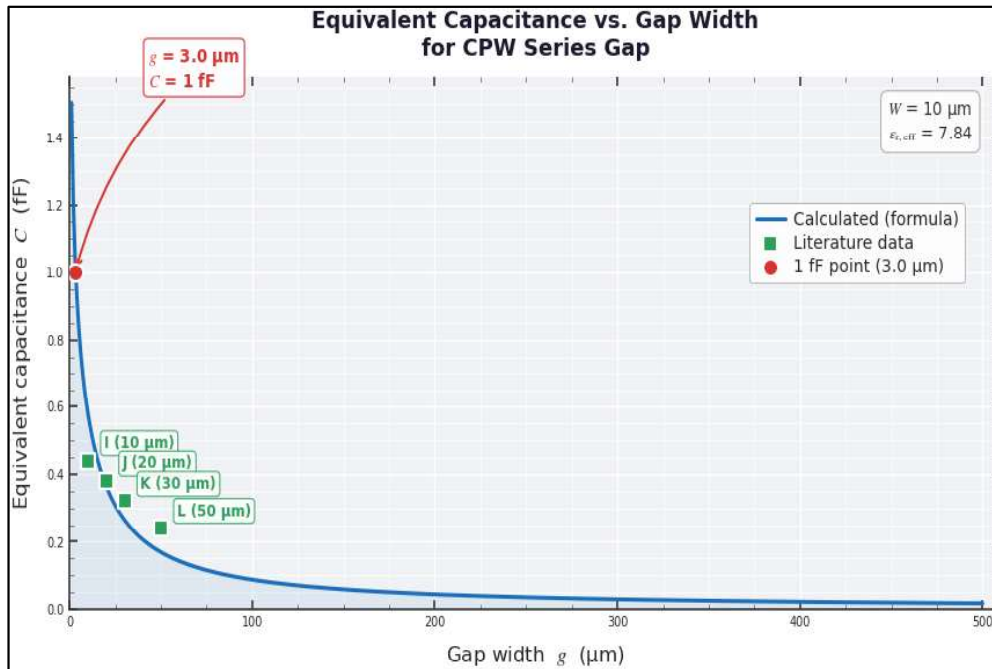


Figure 12. Comparison of modelled capacitance values with literature data for the gap capacitor

Good agreement with literature values. Based on this, the coupling capacitance values used in the cavity design were computed using this model, as listed in the table below. The calculation for the corresponding resonant lengths for the selected frequencies is given in Appendix 4.

Capacitance(fF)	Gap Width (μm)	Resonant Frequency f_0 (THz)	Resonant Length l (μm)
0.26	30	0.3	178.57
0.5	13	0.5	107.14
1	3	0.8	66.69
5	1.14	--	--
10	0.55	--	--
20	0.3	--	--

Table 2. Designed CPW Series Gap Capacitance Values and Corresponding Resonant Parameters

Fabrication of CPW Gap Capacitance for THz Cavities

This section describes coplanar waveguide (CPW) resonators with lithographically defined capacitive coupling gaps that control the coupling strength (Q_{ext}) and electromagnetic boundary conditions. Tuning these gaps adjusts the external loading and photon lifetime, enabling THz resonators for optically driven light–matter interaction, a key component to this project.

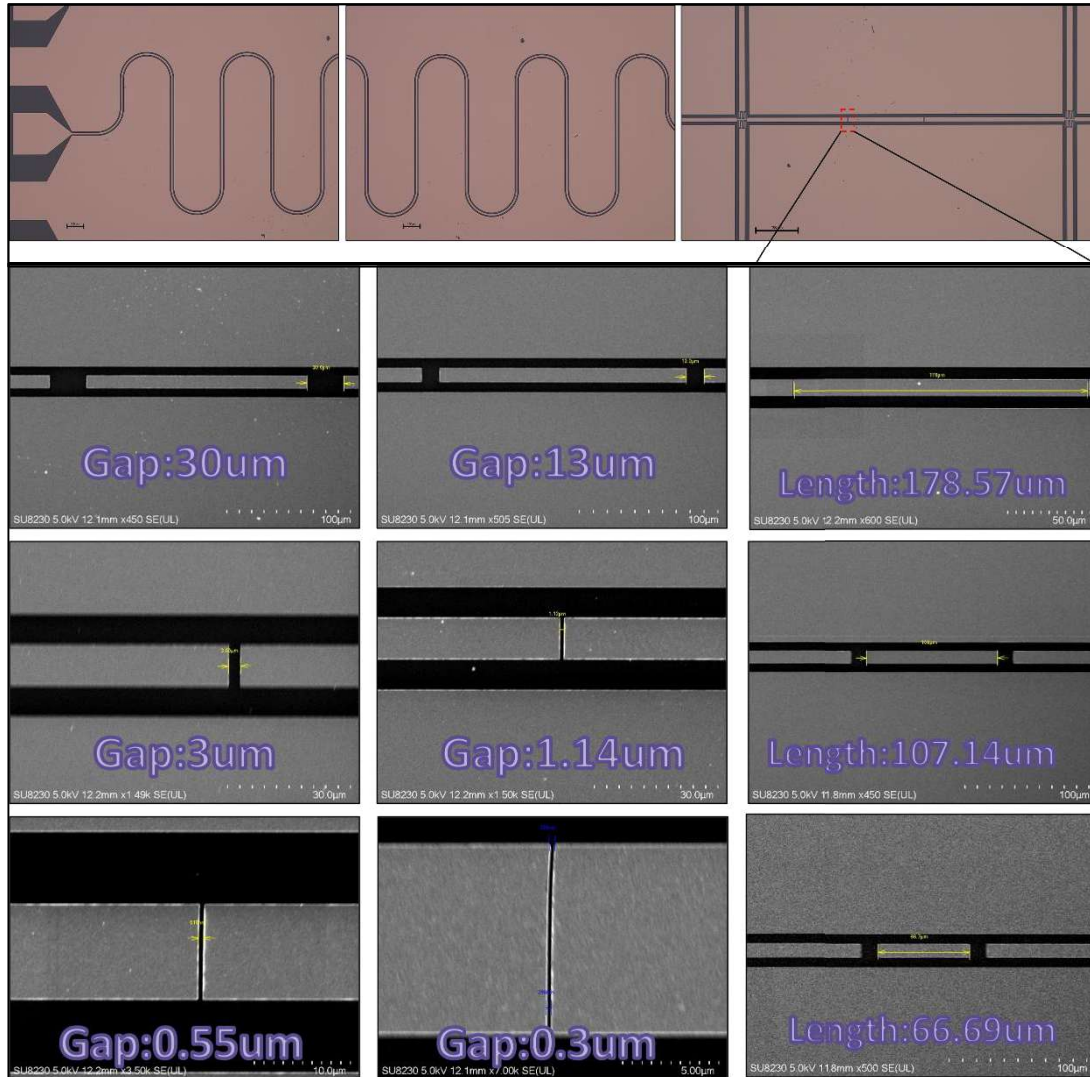


Figure 13. Coplanar waveguide incorporating all designed capacitive coupling gap capacitors

As shown, a representative coplanar waveguide resonator incorporates a capacitive coupling gap between two meander sections, along with finger switches for laser pulse delivery. The coupling gap controls the coupling strength and thus the external loading of the resonator. The remaining images show the completed set of fabricated devices, including successful metal liftoff for gap sizes down to 0.3 μm .

Fabrication of Cavity Trap: In the following section, we discuss the fabrication process guided by simulation results for the cavity trap. In the THz regime, multiple loss mechanisms, including conductor, dielectric, and radiation losses, can become non-negligible, with radiation losses often contributing strongly at higher frequencies [40-41]. A trench is introduced to improve confinement of the electromagnetic fields in free space.

This section also includes the steps taken to realize the cavity trap, from using different recipes for the thick metal mask to conducting different dry etching processes to achieve the target trench profile and depth.

Mask Recipe Development: It started with an existing recipe in our group, using HSQ as the resist and Ni as the mask material. However, the recipe did not work well with a long HF lift-off time of 5 min and a development time of 1 min, as shown below in figure 15. So, we reduced the lift-off time from 5 min to 1 min as a starting point and increased the development time from 1 min to 1 min 30 s. This worked well for the trench design we had: 2 μm , 5 μm , and 10 μm wide, and 1000 μm long, as shown below.

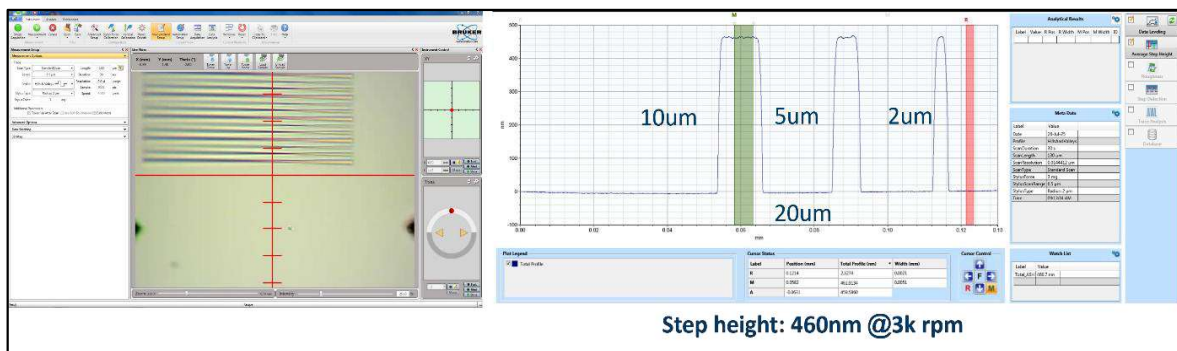


Figure 14. Dektak profilometer trace of all trenches 2um, 5um and 10um after resist development step. As shown, the Dektak profilometer trace confirms the expected trench height and width after optimized development for 1 min 30 s.

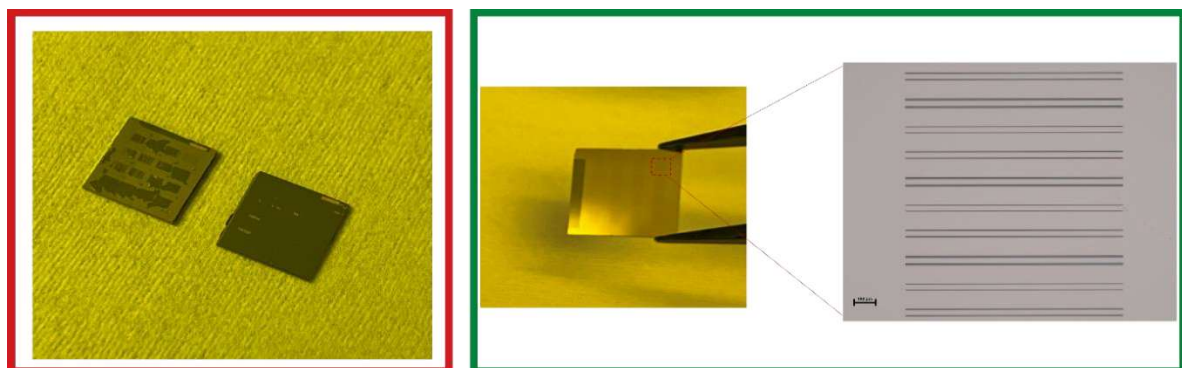


Figure 15. Failed trials compared to a successful one after HF liftoff

As shown, the images on the left show failed trials, while the image on the right shows a successful result after a 1 min optimized HF liftoff of the Ni, with a close-up microscopic view of the trenches.